

# CHAPTER 6 Computer Networks and Internet

- Describe local and wide area networks (LANs and WANs).
- Distinguish an internet from the Internet.
- Describe the TCP/IP protocol suite as the network model in the Internet.
- Define the layers in the TCP/IP protocol suite and their relationship.
- Describe some application at the application layer.
- Describe the services provided by the transport-layer protocols.
- Describe the services provided by the network-layer protocol.
- Describe different protocols used at the data-link layer.
- Describe the duties of the physical layer.
- Describe different transmission media used in networking.

## 6.1 INTRODUCTION

- A **network** is a combination of hardware and software that sends data from one location to another.
- The hardware consists of the physical equipment that carries signals from one point in the network to another.
- The software consists of instructions that make the services that we expect from a network possible.

## 6.1.1 Network criteria

- A network must be able to meet a number of criteria.
- The most important of these are performance, reliability, and security.
  - **Performance** can be measured in many ways, including transit time and response time.
  - **Reliability** is measured by the frequency of failure, the time it takes to recover from a failure, and the network's robustness in a catastrophe.
  - **Network security** issues include protecting data from unauthorized access, damage, and change, and implementing policies and procedures for recovery from breaches and data losses.

## 6.1.2 Physical structures

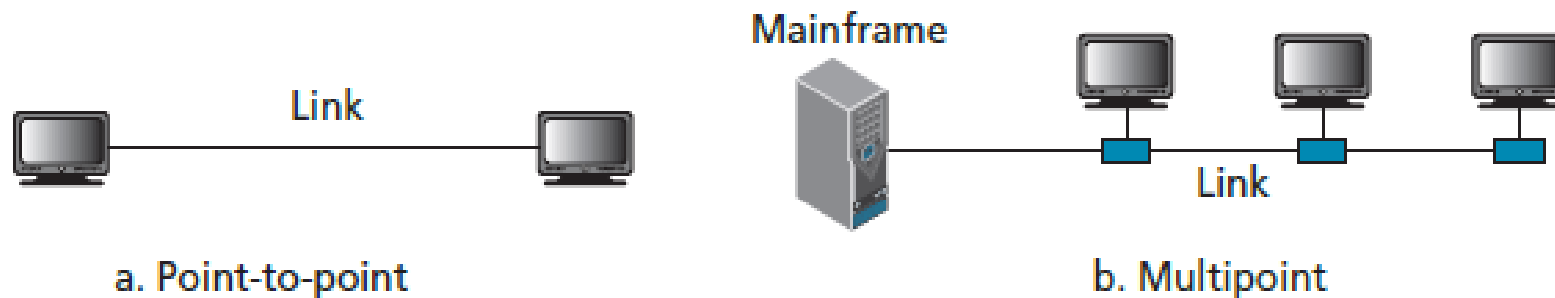
Before discussing networks, we need to define some network attributes.

- **Types of Connection:** A network consists of two or more devices connected through **links**.
- A link is a communications pathway that transfers data from one device to another. There are two possible types of connections: **point-to-point** and **multipoint**.

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Figure 6.1 *Types of connections: point-to-point and multipoint*

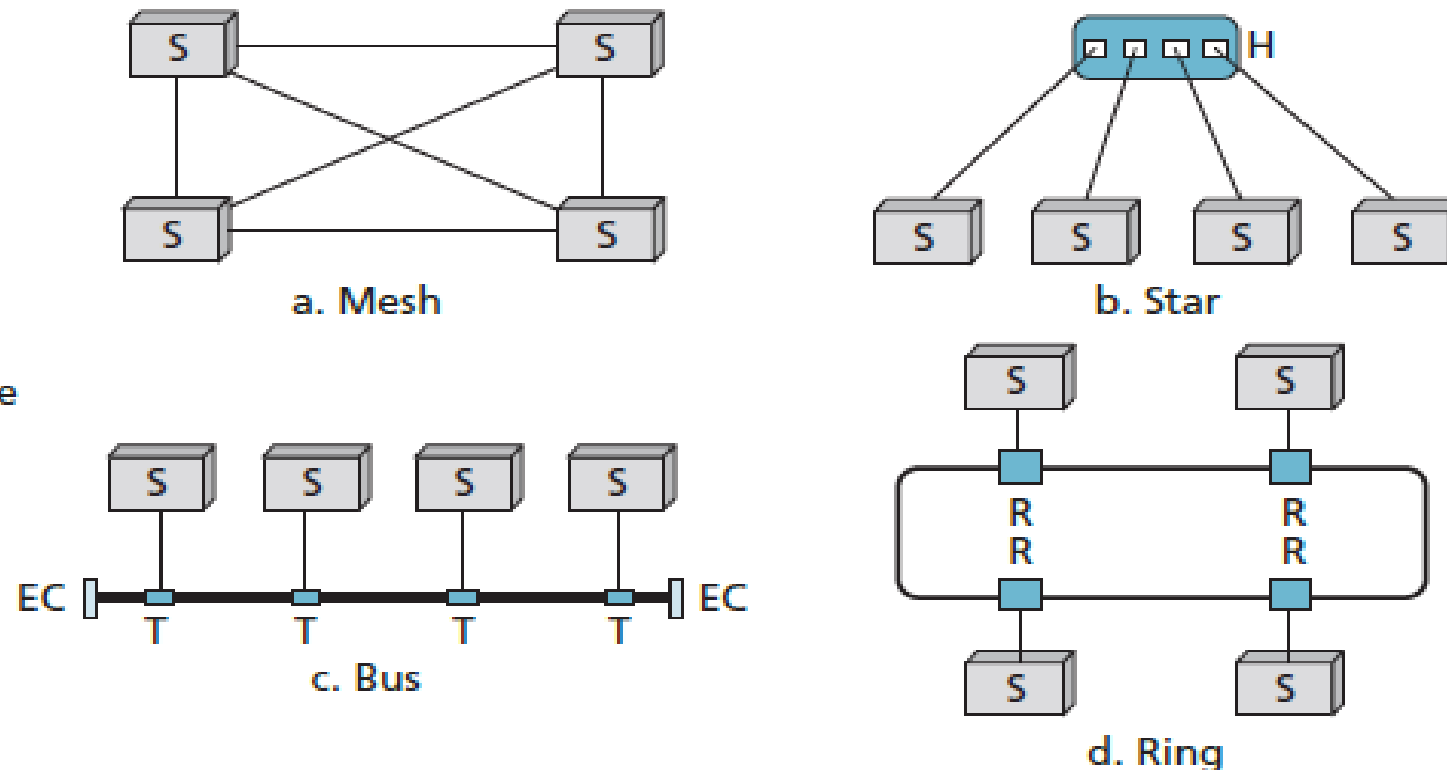
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- The term **physical topology** refers to the way in which a network is laid out physically.
- There are four basic topologies possible: **mesh**, **star**, **bus**, and **ring**.

Figure 6.2 Four physical topologies

Legend:  
S: Station  
R: Repeater  
H: Hub  
T: Tab  
EC: End Cable



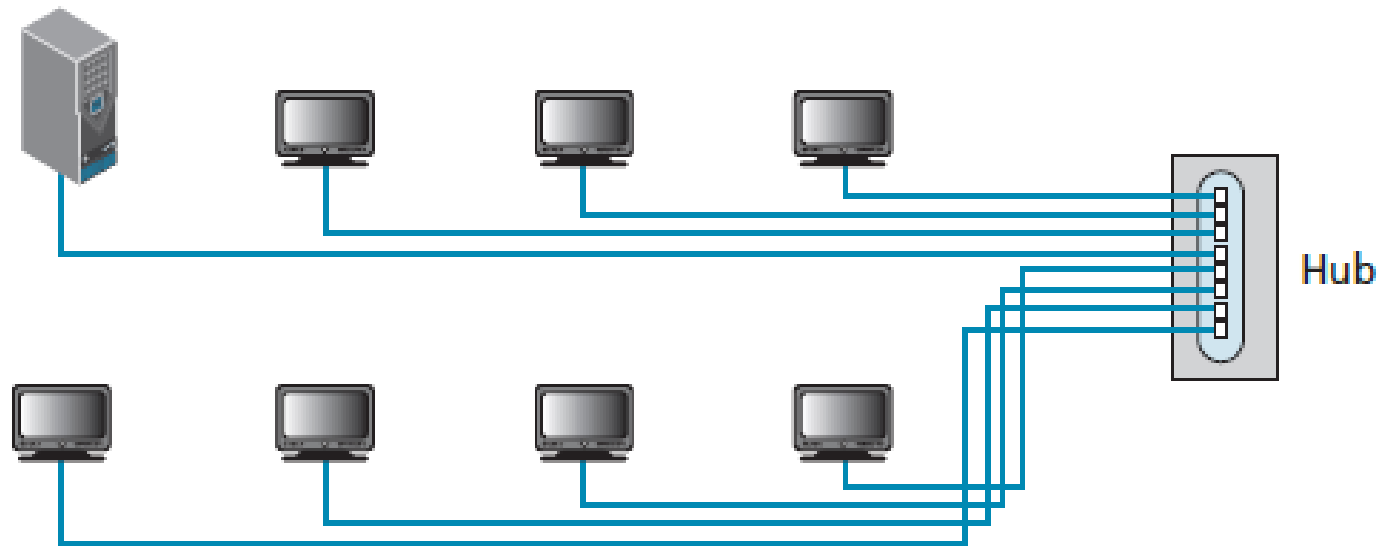
# Property Comparison

| Feature              | Mesh             | Star                | Bus                   | Ring                                       |
|----------------------|------------------|---------------------|-----------------------|--------------------------------------------|
| Cable Length         | Very high        | Moderate            | Low                   | Moderate                                   |
| Cost                 | Very expensive   | Moderate            | Cheap                 | Moderate                                   |
| Reliability          | Very high        | High                | Low                   | Moderate                                   |
| Ease of Installation | Complex          | Easy                | Easy                  | Moderate                                   |
| Failure Impact       | None (redundant) | Hub failure = total | Cable failure = total | Node failure = total<br>(unless dual ring) |

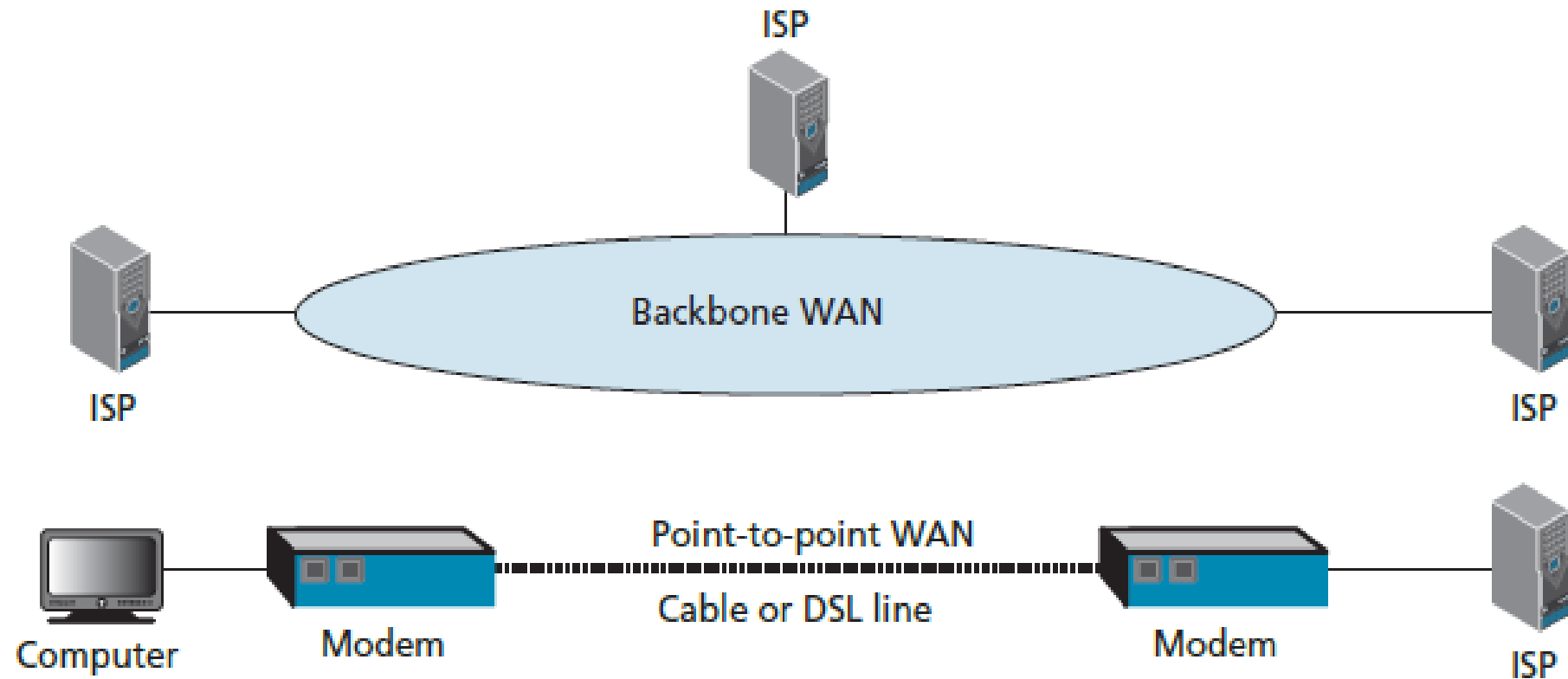
## 6.1.3 Categories of networks

- Today networks can be divided into three broad categories: **local-area networks (LANs)**, **wide-area networks (WANs)**, and **metropolitan area networks (MANs)**.

**Figure 6.3** *An isolated LAN connecting eight computers to a hub*



**Figure 6.4** *A point-to-point WAN and a backbone WAN*

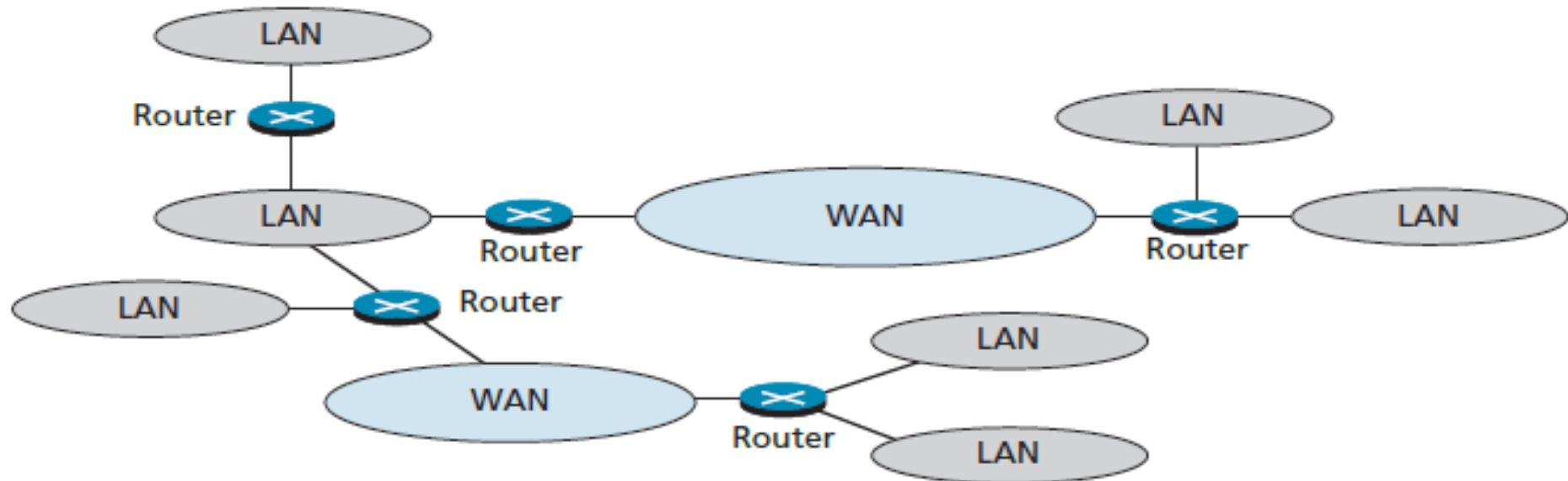




## 6.1.4 An internet

- Today, it is very rare to see a network in isolation: networks are connected to one another.
- When two or more networks are connected, they become an **internetwork**, or an **internet** (lowercase “i”).

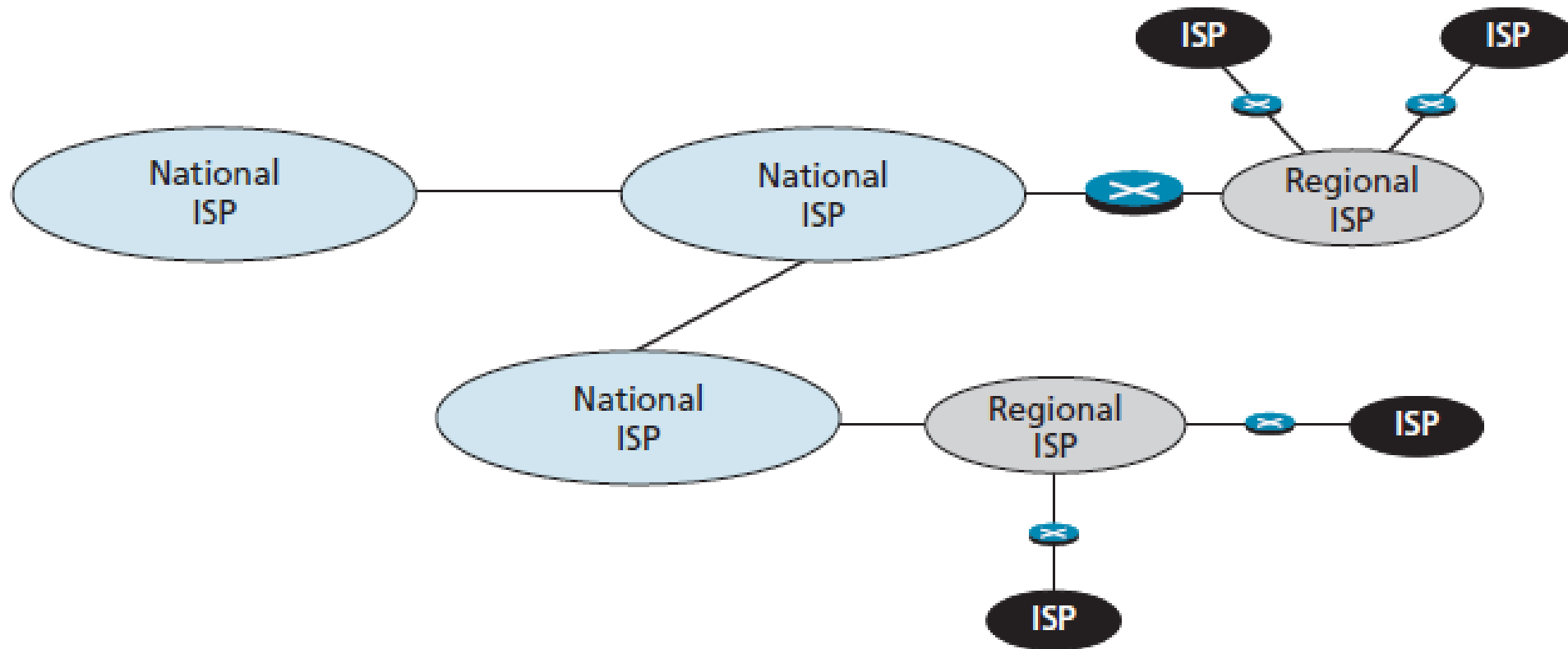
Figure 6.5 *An internet made of WANs, LANs, and routers*



## 6.1.5 The Internet

- The most notable internet is the **Internet** (uppercase “I”), a collaboration of hundreds of thousands of interconnected networks.
- Private individuals, as well as various organizations such as government agencies, schools, research facilities, corporations, and libraries in more than 100 countries use the Internet.
- Millions of people are users.
- It is difficult to give an accurate representation of the Internet, because it is continually changing.
- Today most end users who want an Internet connection use the services of **Internet service providers (ISPs)**.

**Figure 6.6** *Hierarchical organization of the internet*



# Taiwan主要 ISP（全國性或大型業者）

| 名稱                                                              | 特點／備註                            |
|-----------------------------------------------------------------|----------------------------------|
| 中華電信 (Chunghwa Telecom / CHT / HiNet)                           | 傳統上是臺灣最大的電信／網路業者，許多其他業者租用它的基礎設施。 |
| 台灣大哥大 / Taiwan Mobile                                           | 也提供有線／無線網路服務，是主要電信業者之一。          |
| 遠傳電信 (Far EasTone / FET / FETNET)                               | 提供電信和網路服務。                       |
| 數位聯合 (Digital United / DUNet / SeedNet 所屬)                      | 在大都市和校園網路中相當活躍。                  |
| 新世紀資通 (New Century InfoComm Tech Co., Ltd.)                     | 在臺灣有一定規模的網路服務提供者。                |
| 台灣固網 (Taiwan Fixed Network, Telco and Network Service Provider) | 在 IP 分配與網路骨幹中也有角色。               |
| 學術網 TANet (Taiwan Academic Network)                             | 提供大專院校、研究機構的網路互聯，是教育／研究網路骨幹。     |

## 6.2 TCP/IP PROTOCOL SUITE

- To divide the services required to perform a task, the Internet has created a set of rules called protocols. These allow different local and wide area networks, using different technologies, to be connected together and carry a message from one point to another.
- The set, or suite, of protocols that controls the Internet today is referred to as the **TCP/IP protocol suite**.
- The abbreviations (TCP and IP) will become clear as we explain different protocols.

- The original TCP/IP protocol suite was defined as having four layers: host-to-network (or link), internet (network), transport, and application. However, the TCP/IP protocol suite today is normally considered as a five-layer model, as shown in Figure 6.7.

**Figure 6.7** *The TCP/IP protocol suite*

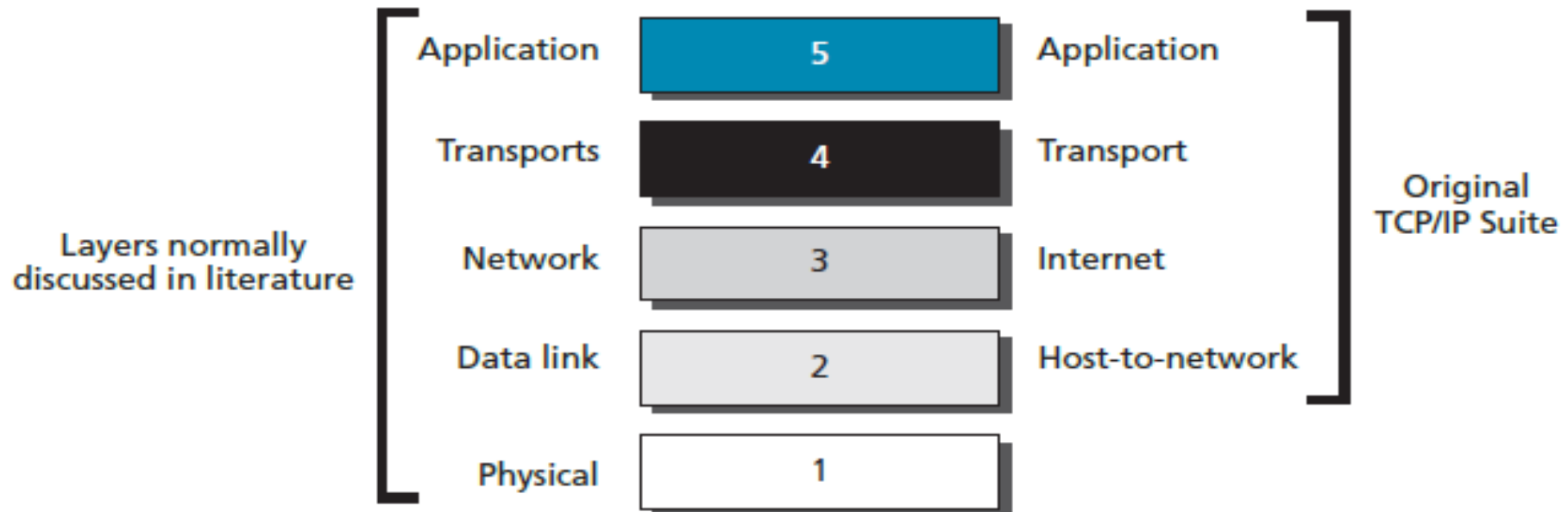
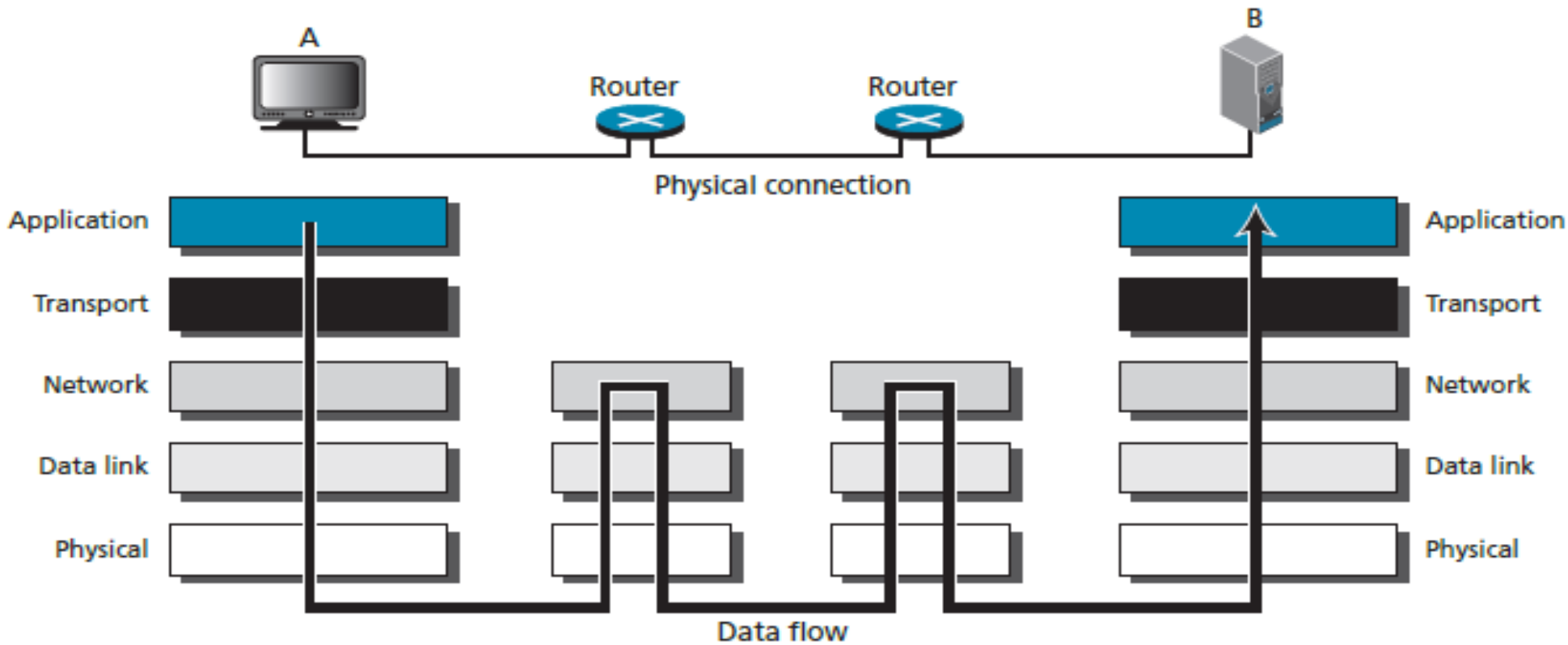


Figure 6.8 shows the layers involved when a message is sent from device A to device B. As the message travels from A to B, it may pass through many routers. Routers use only the first three layers.

Figure 6.8 The interaction between layers in the TCP/IP protocol suite



## 6.3 LAYERS

- This section briefly describes the function of each **layer** in the TCP/IP protocol suite. We show how a message travels through the different layers until it reaches the physical layer and is sent by the transmission media.

### 6.3.1 Application layer

- The application layer enables a user, whether human or software, to access the network.
- It provides support for services such as electronic mail, remote file access and transfer, browsing the World Wide Web, and so on.

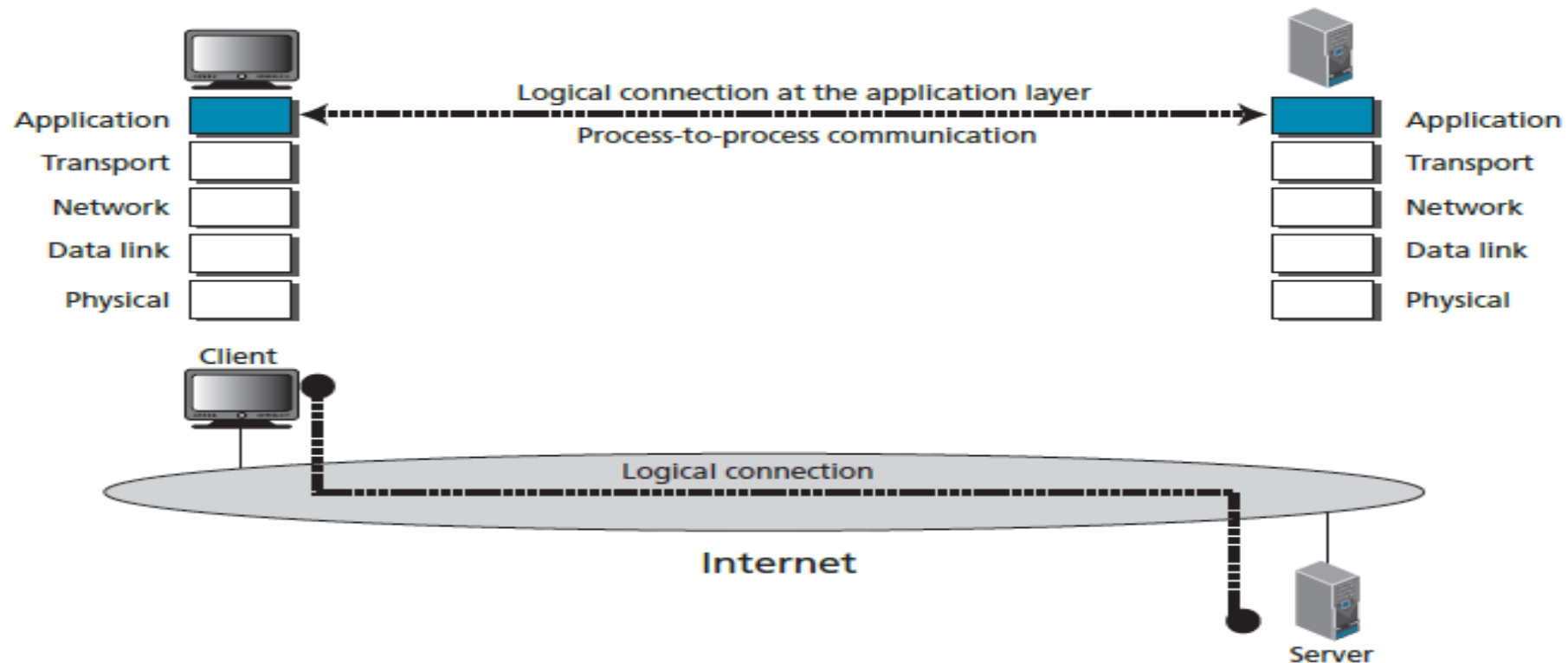
The application layer is responsible for providing services to the user.



# Client-server architecture

- Although there are two architectures (designs) that allow two application programs, running on two remote computers, to communicate with each other, **client-server architecture** is more common.

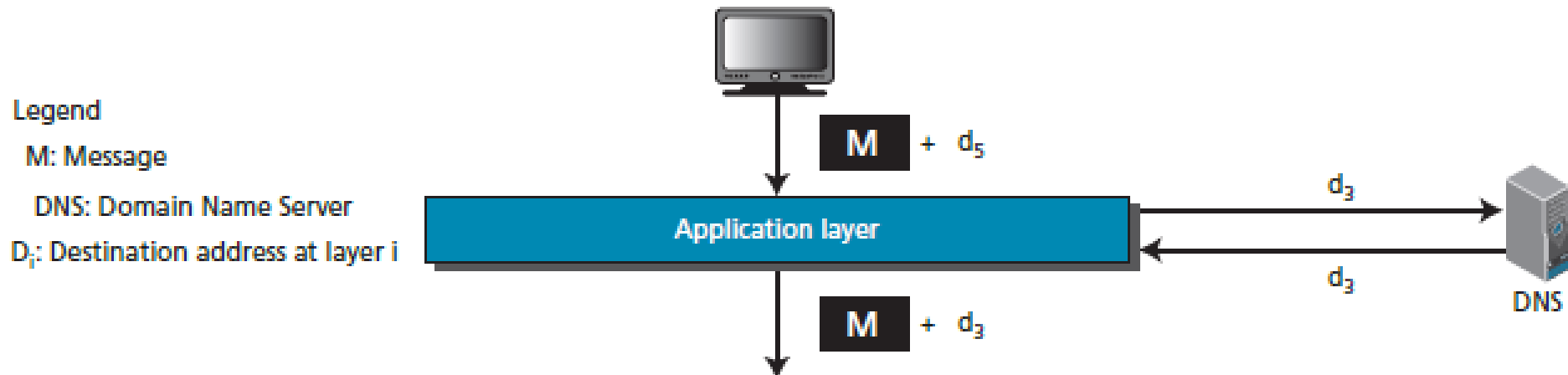
**Figure 6.9** *Communication at the application layer*



# Application-layer address

- When a client needs to send a request to a server, it needs the server application-layer address. For example, to identify one particular site, the client uses a Uniform Resource Locator (URL).
- As we will see later, the server application-layer address is not used for delivery of messages; it only helps the client to find the actual address of the server computer.

**Figure 6.10** *Addresses at the application layer*

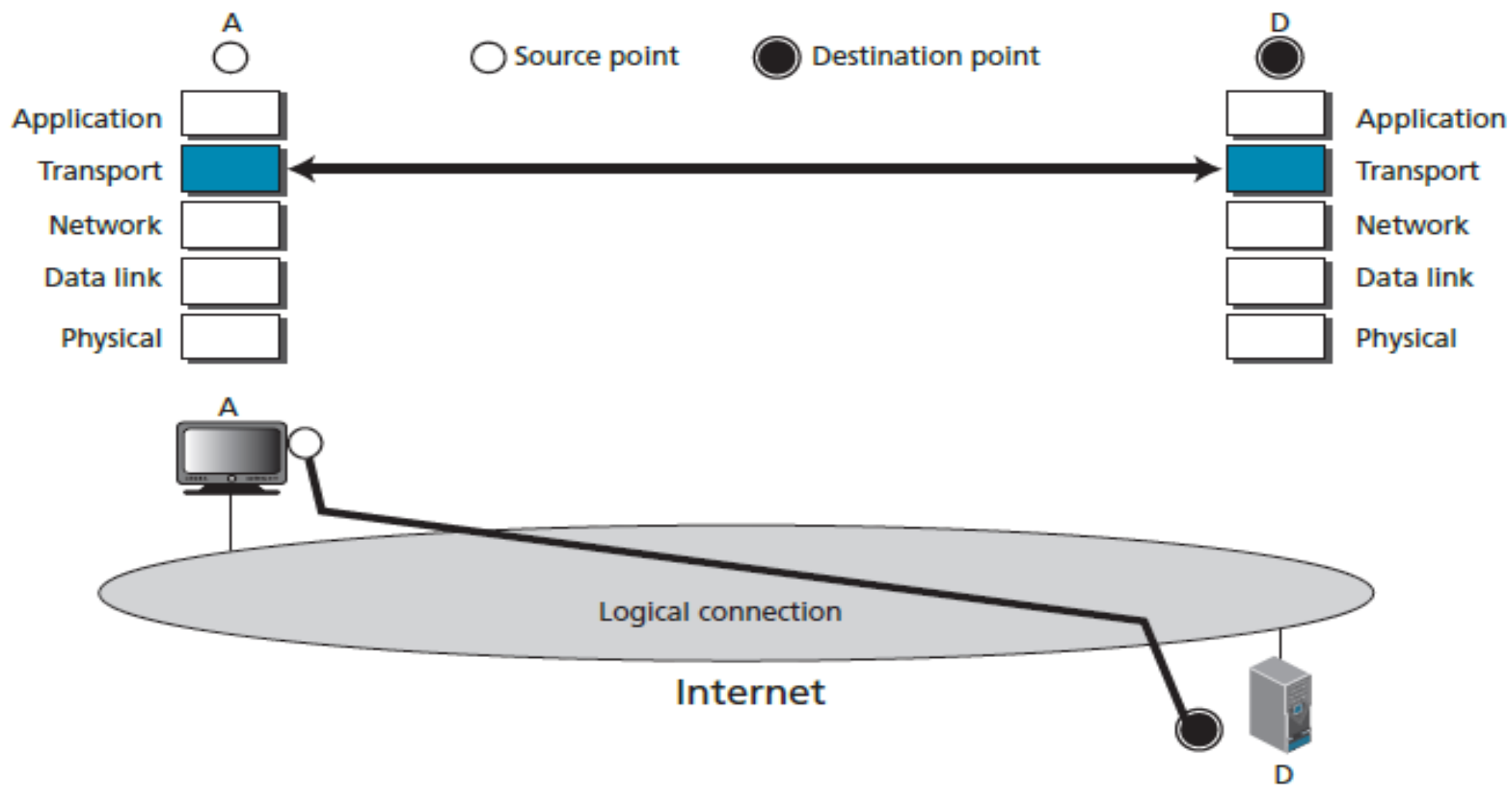


## 6.3.2 Transport layer

- The transport layer is responsible for process-to-process delivery of the entire message: logical communication is created between the transport layer of the client and the server computer.
- In other words, although physical communication is between two physical layer (through many possible links and routers), the two application layers consider the transport layer as the agent that takes responsibility for delivering the messages.

The transport layer is responsible for the logical delivery of a message between client and server processes.

**Figure 6.11** *Communication at the transport layer*



# Transport-layer addresses (port numbers)

- The server computer may be running several processes at the same time, for example an FTP server process and an HTTP server process.
- When the message arrives at the server, it must be directed to the correct process. We need another address for server process identification, called a port number.

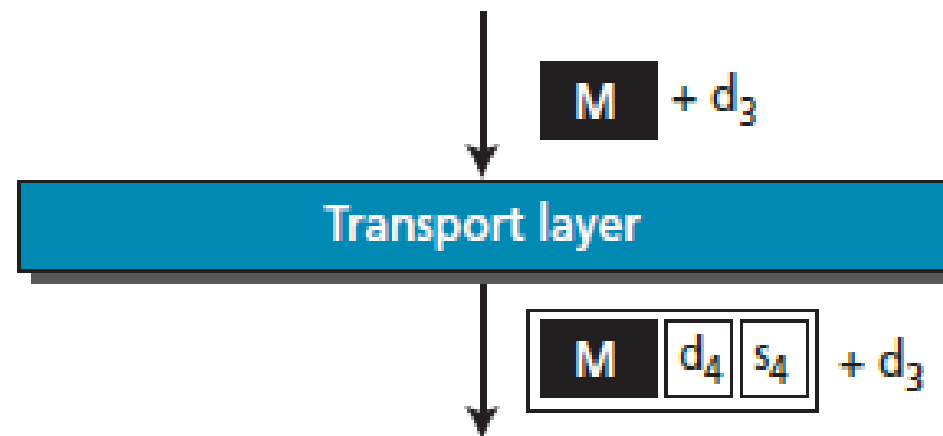
Figure 6.12 *Addresses at the transport layer*

Legend

M: Message

$d_i$ : Destination address at layer  $i$

$s_i$ : Source address at layer  $i$



# Transport-layer protocols

- During the life of the TCP/IP protocol suite three transport layer protocols have been designed: **UDP**, **TCP**, and **SCTP**.
- The **User Datagram Protocol (UDP)** is the simplest of all three protocols. UDP does multiplexing and de-multiplexing. It also does a type of error control by adding a checksum to the packet.
- **Transmission Control Protocol (TCP)** is a protocol that supports all the duties of a transport layer. However, it is not as fast and as efficient as UDP. TCP uses sequence numbers, acknowledgment numbers, and checksums. It also uses buffers at the sender's site. This combination of provisions provides multiplexing, de-multiplexing, flow control, congestion control, and error control.
- **Stream Control Transmission Protocol (SCTP)** is a new protocol that is designed for new services expected from the Internet, such as Internet telephony and video streaming. This protocol combines the advantages of both UDP and TCP. Like UDP, it is suitable for real-time transmission of audio and video, but like TCP, it provides error and flow control.

## 6.3.3 The network layer

- The network layer is responsible for the source-to-destination (computer-to-computer or host-to-host) delivery of a packet, possibly across multiple networks (links).
- The network layer ensures that each packet gets from its point of origin to its final destination.

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

## Network-layer addresses

- The packet traveling from the client to the server and the packet returning from the server need network-layer address.
- The server address is provided by the server, as discussed above, while the client address is known by the client computer.

Figure 6.13 *Addresses at the network layer*

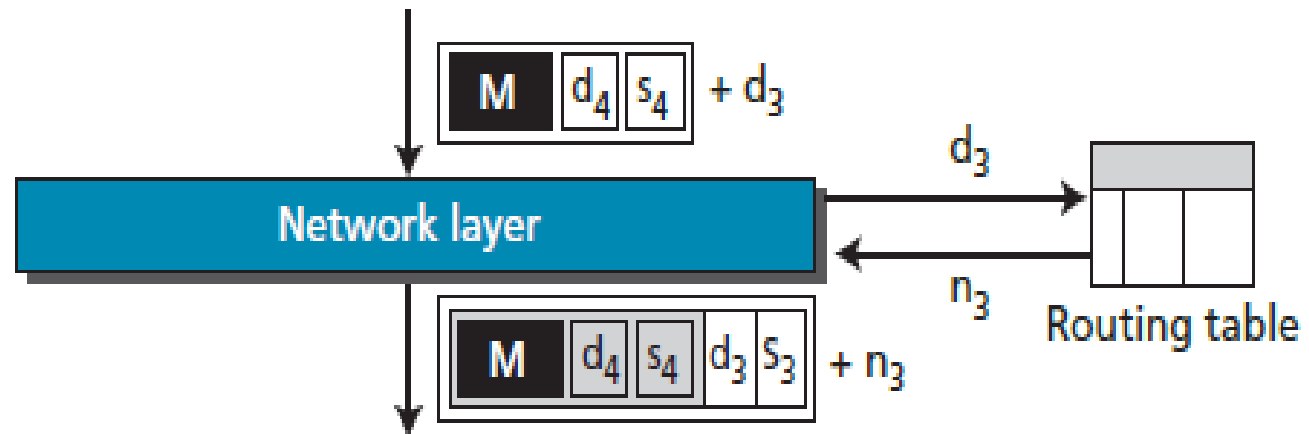
### Legend

M: Message

$d_i$ : Destination address at layer  $i$

$s_i$ : Source address at layer  $i$

$n_i$ : Next-hop address at layer  $i$

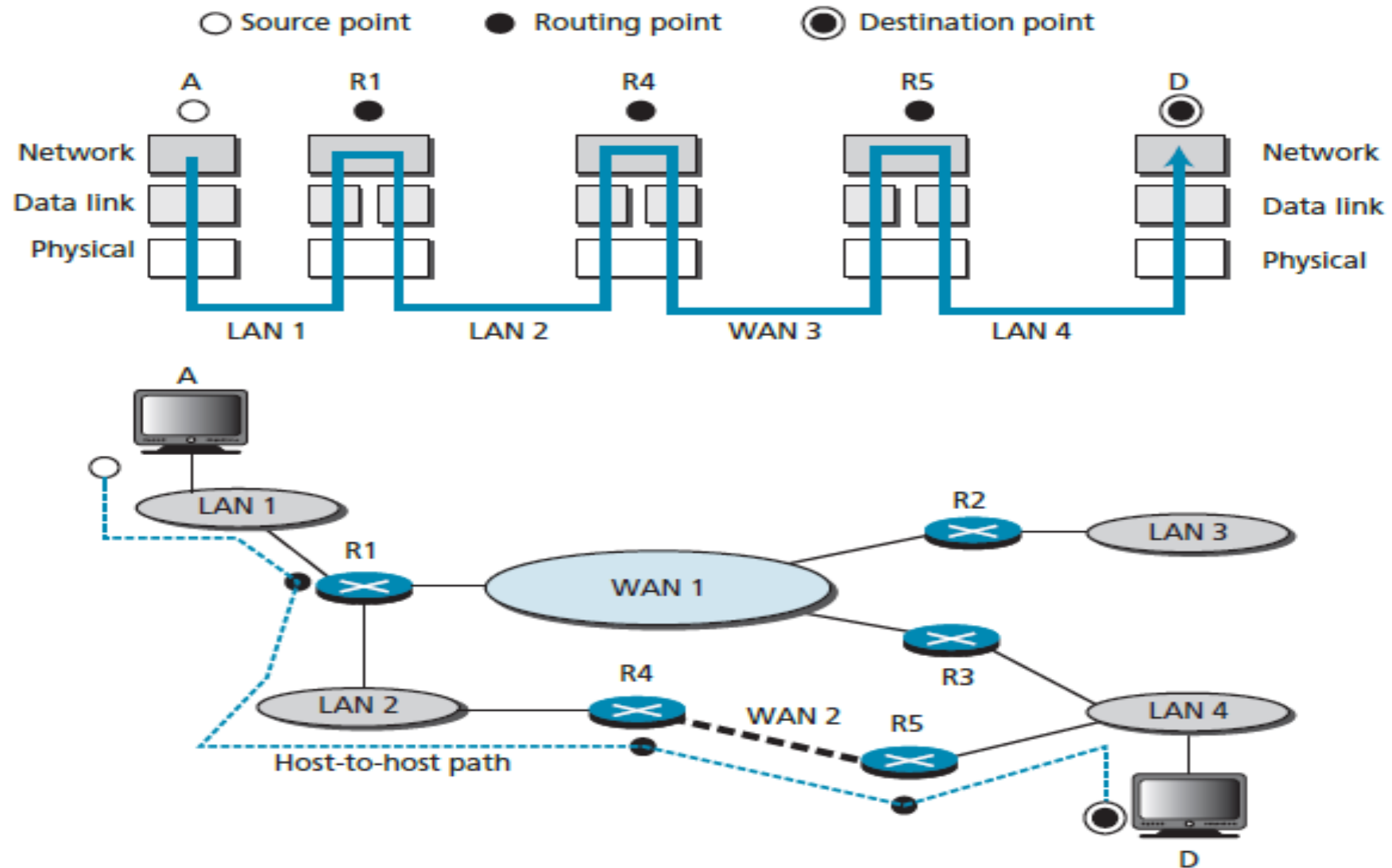




# Routing

- The network layer has a specific duty: routing.
- Routing means determination of the partial or total path of a packet.
- As the Internet is a collection of networks (LANs, WANs, and MANs), the delivery of a packet from its source to its destination may be a combination of several deliveries: a source-to-router delivery, several router-to-router delivery, and finally a router-to-destination delivery.

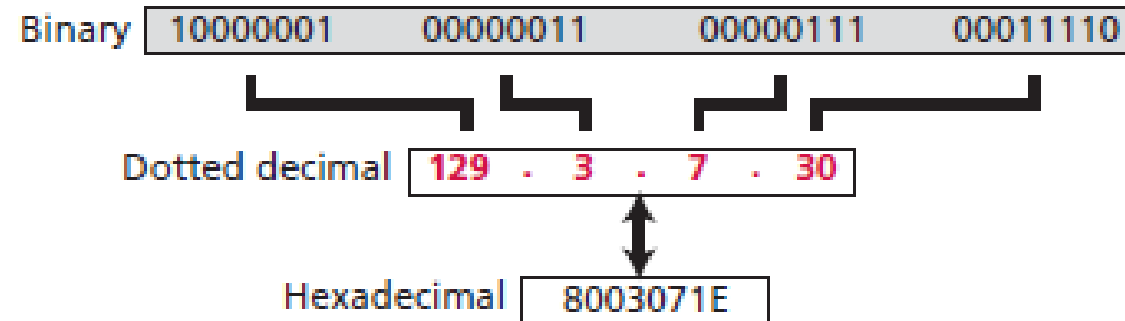
Figure 6.14 Routing at the network layer



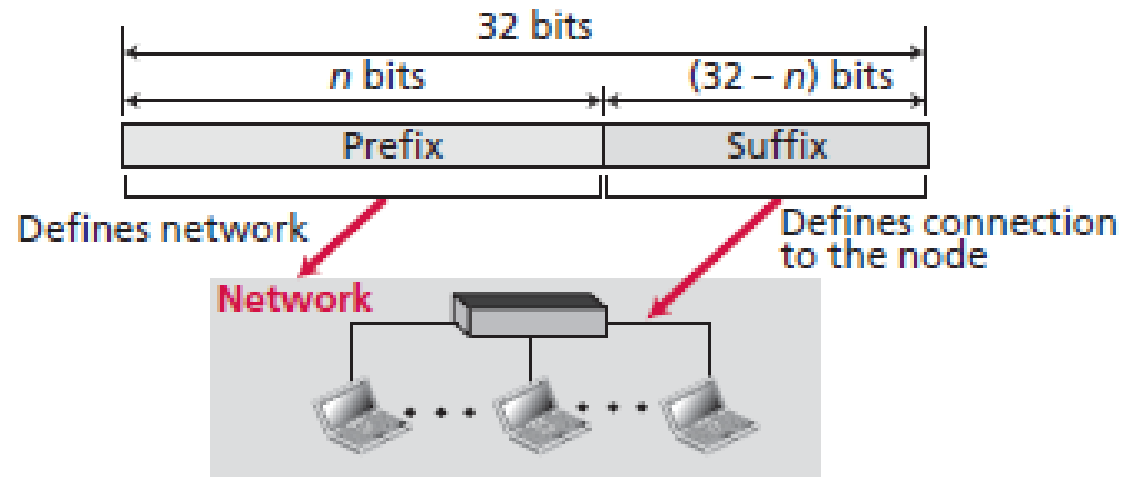
# Network-layer protocols

- The TCP/IP protocol suite supports one main protocol (IP) and several auxiliary protocols to help IP to perform its duties.
- In the TCP/IP protocol suite, the main protocol at the network layer is **Internet Protocol (IP)**. The current version is IPv4 (version 4) although IPv6 (version 6) is also in use, although not ubiquitously.
- **IPv4** is responsible for the delivery of a packet from the source computer to the destination computer. Every computer and router in the world is identified by a **32-bit** IP address. The notation divides the 32-bit address into four 8-bit sections and writes each section as a decimal number between 0 and 255 with three dots separating the sections.
  - **IPv4** address **00001010 00011001 10101100 00001111** is written as **10.25.172.15**
- **IPv6** uses **128 bits** to define any device connected to the Internet.
  - **IPv6** address **FEF7:5638:FF23:A2B5:BC21:0234:7256:1345**

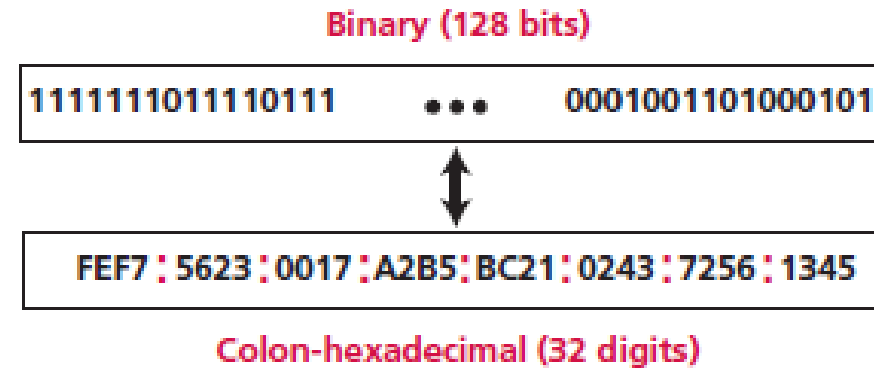
## Address notation



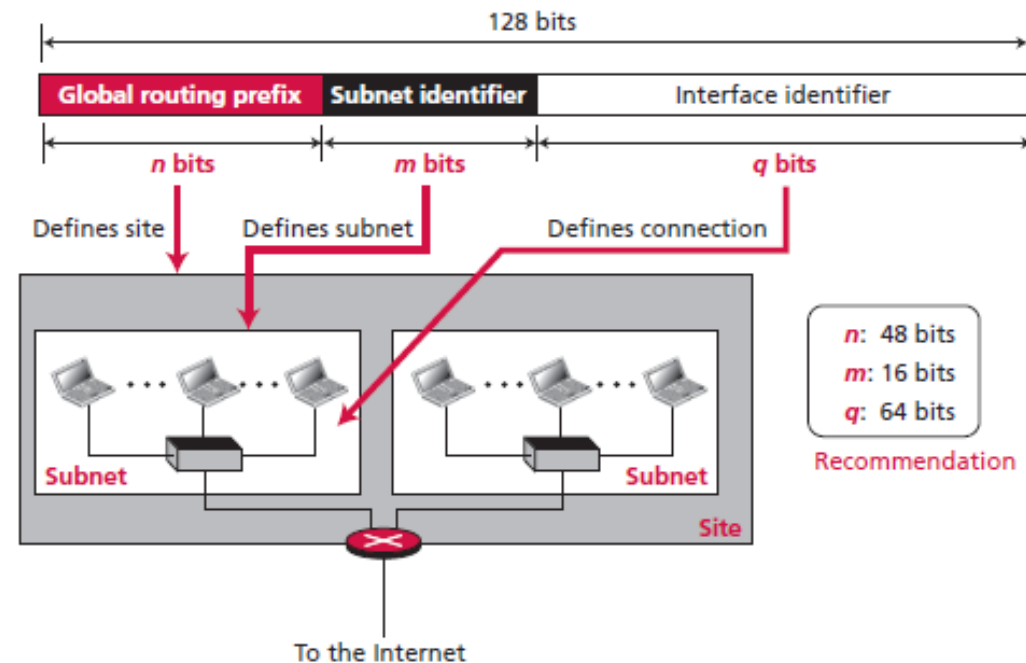
## Hierarchy in IPv4 addressing



## IPv6 address notation



## Hierarchy in IPv6 addressing

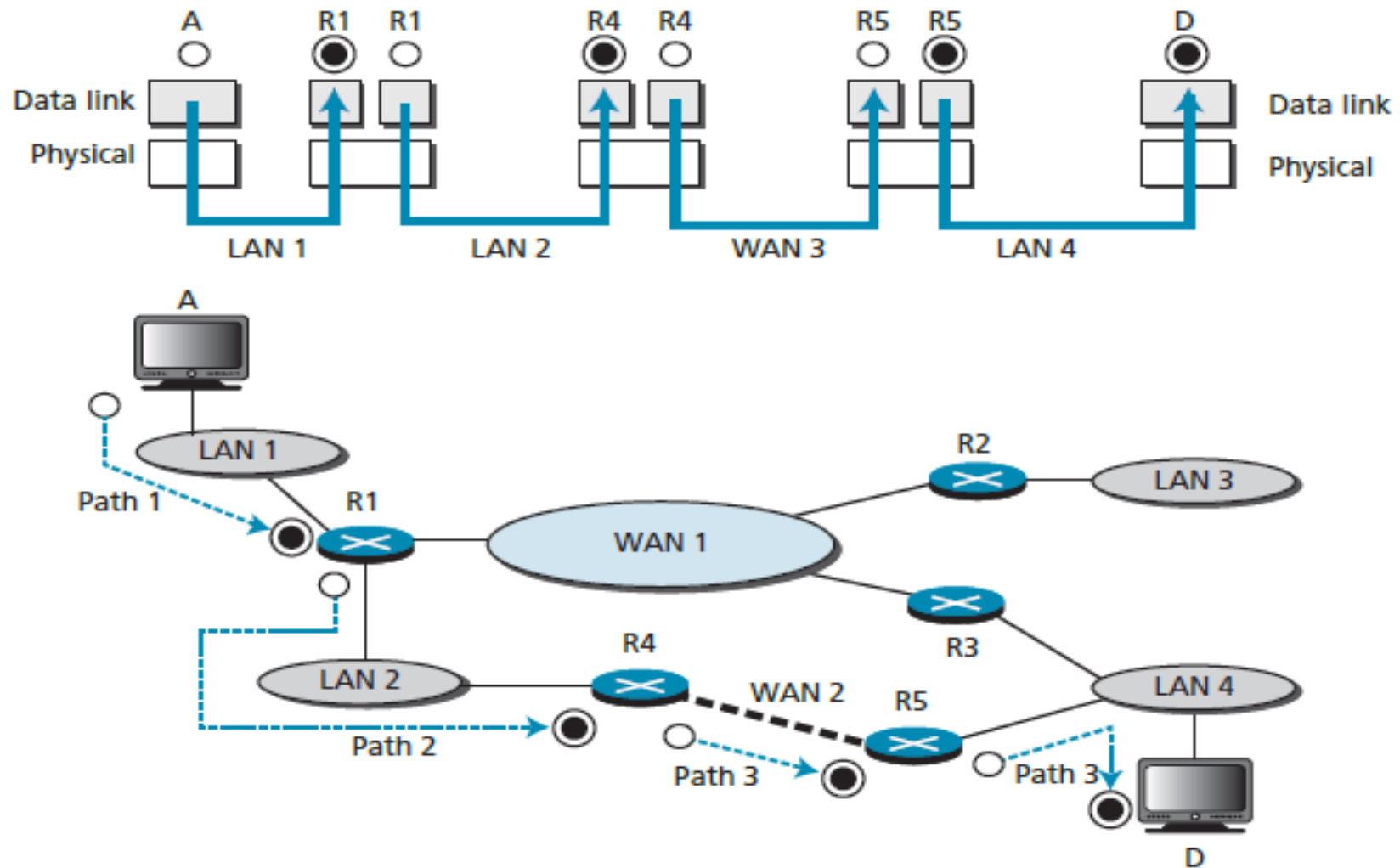


## 6.3.4 Data link layer

- As we saw in the previous section, the network layer packet may pass through several routers in its journey from its source to its destination.
- Carrying the packet from one node to another (where a node can be a computer or a router) is the responsibility of the data link layer.

The data link layer is responsible for node-to-node delivery of frames.

**Figure 6.15** *Communication at the data-link layer*



## Data-link layer addresses

- Two questions that come to mind are how computer A knows the data-link layer address of router  $R_1$ , or router  $R_1$  knows the data-link layer address of router  $R_4$ .
- A device can find the data-link address of another device either **statically** or **dynamically**.

Figure 6.16 Addresses at the data link layer

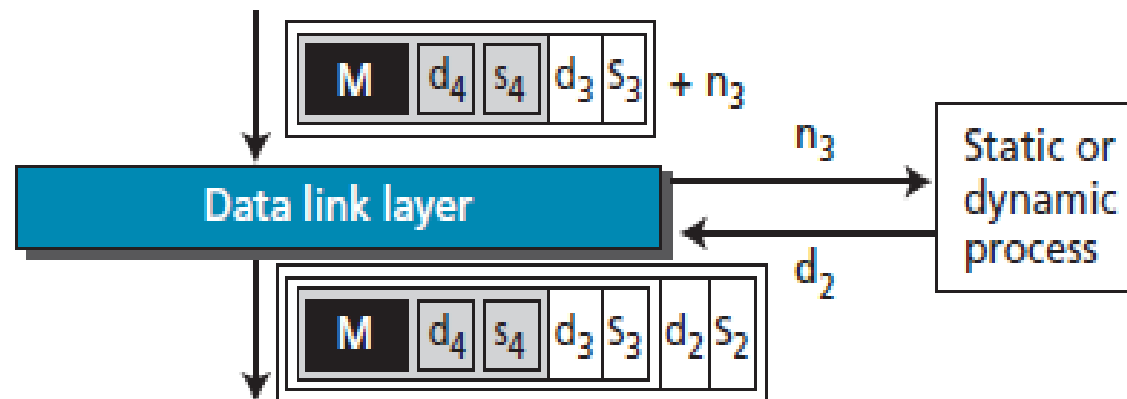
### Legend

M: Message

$d_i$ : Destination address at layer  $i$

$s_i$ : Source address at layer  $i$

$n_i$ : Next-hop address at layer  $i$





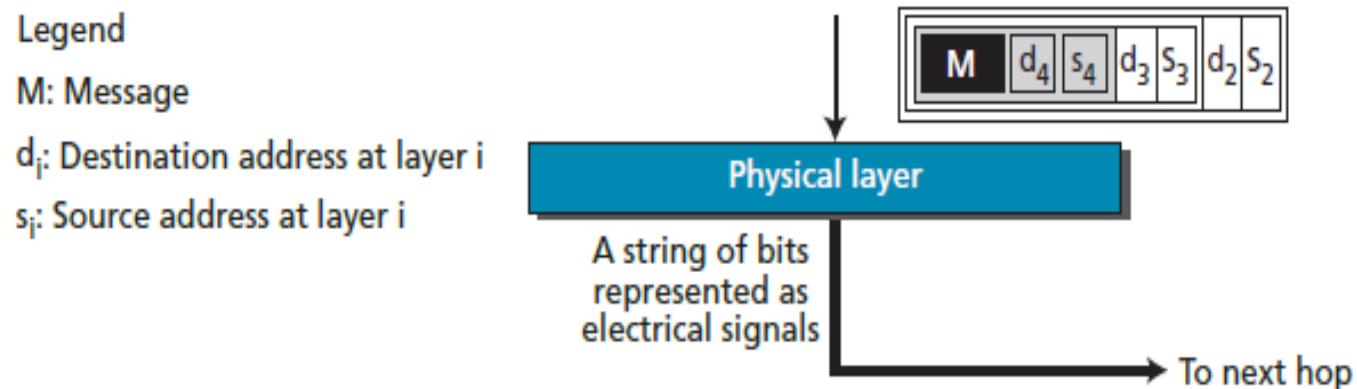
- Unlike IP addresses, addresses at the data-link layer cannot be universal.
- Each data link protocol may have a different address format and size.
- The Ethernet protocol, the most prevalent local area network in use today, uses a 48-bit address, which is normally written in hexadecimal format (grouped in six sections, each with two hexadecimal digits) as shown below:

07:01:02:11:2C:5B

## 6.3.5 Physical layer

- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- Although the data link layer is responsible for moving a **frame** from one node to another, the physical layer is responsible for moving the **individual bits** that make up the frame to the next node. In other words, the unit of transfer in the data link layer is a frame, while the unit of transfer in the physical layer is a bit.

Figure 6.17 Duty of the physical layer

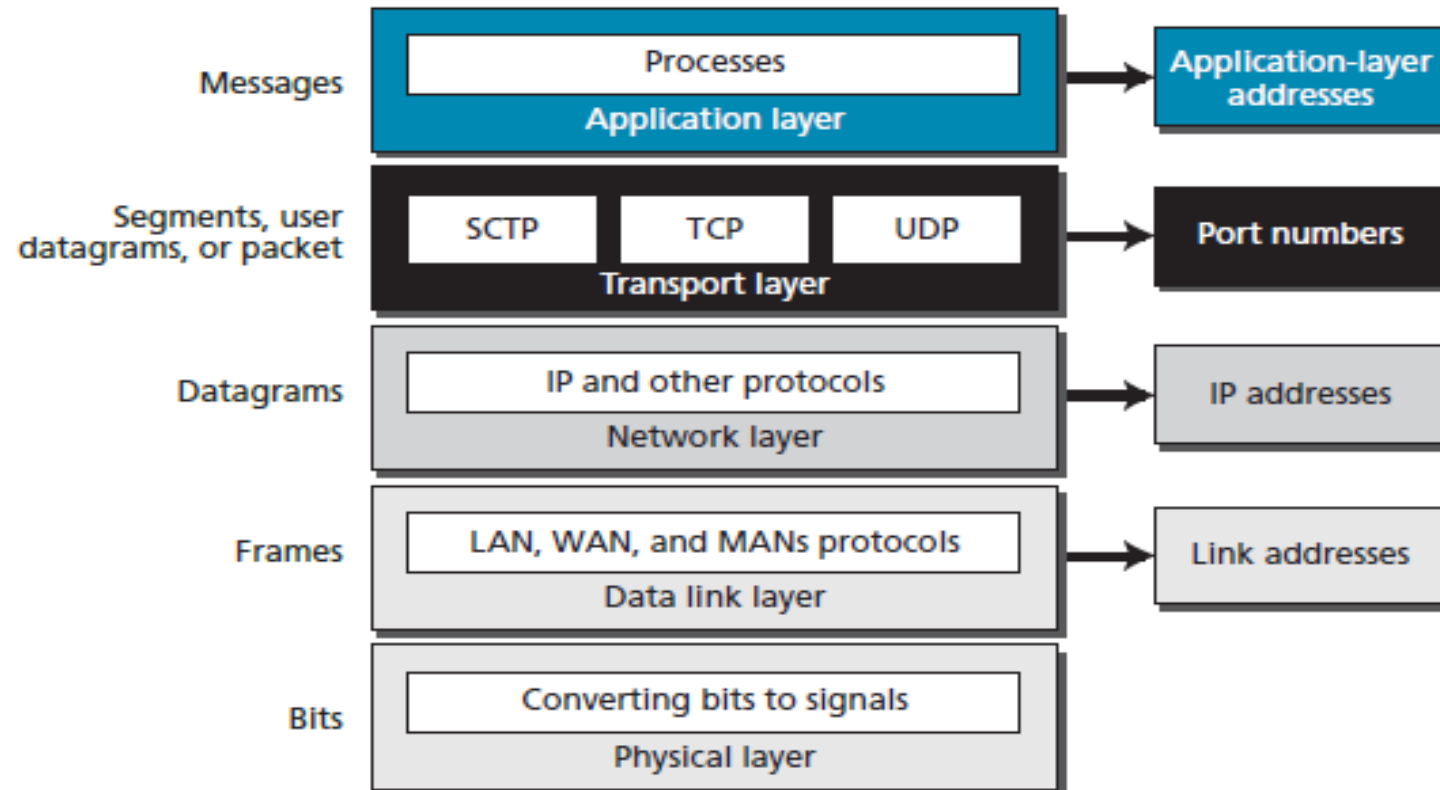


The physical layer is responsible for node-to-node delivery of bits

## 6.3.6 Summary of layers

Figure 6.18 summarizes the duties of each layer in the TCP/IP protocol and the addresses involved in each layer.

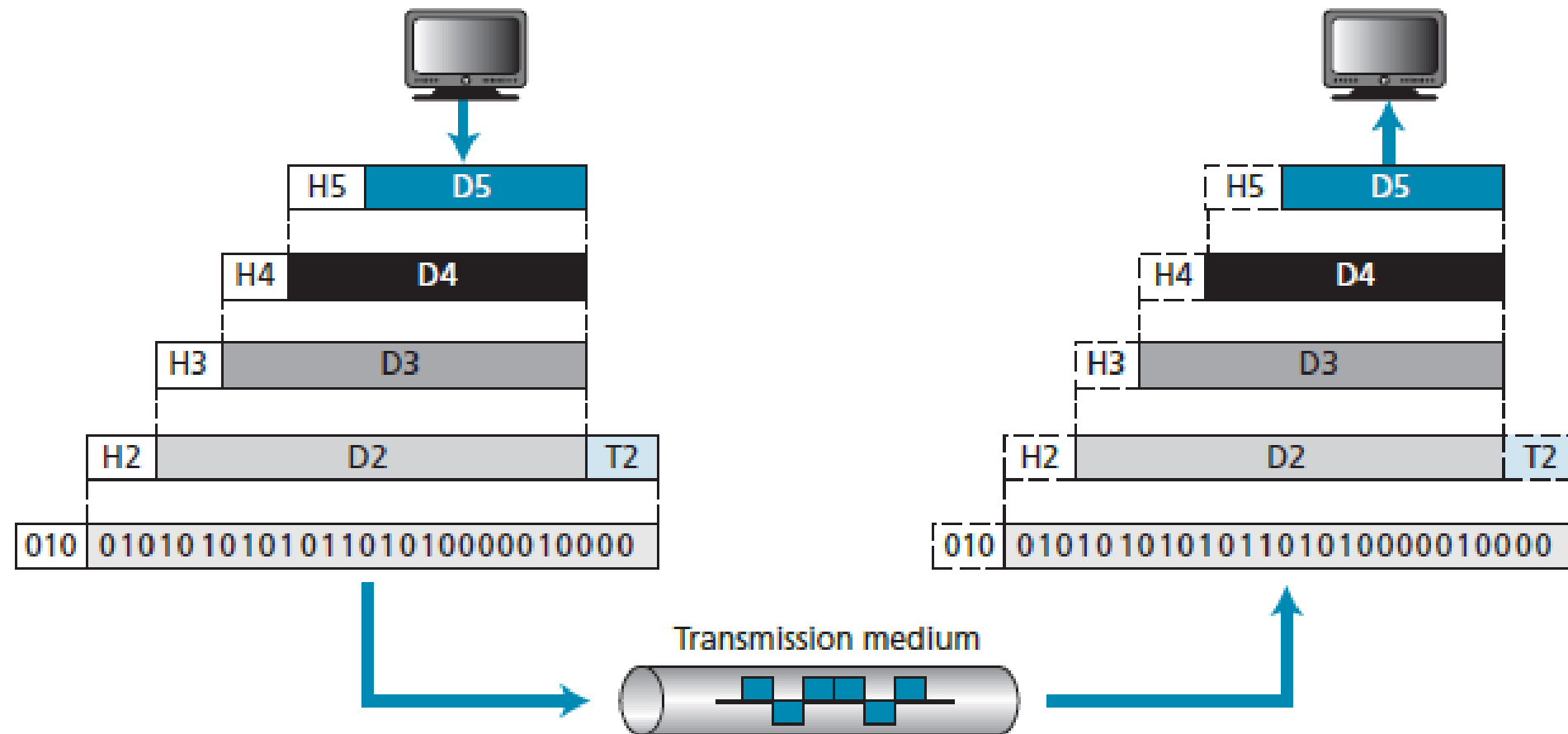
**Figure 6.18** *Four level of addressing in the internet*



# Summary of layers

| Layer / Name                            | Major Protocols / Examples             | Main Function                                                                        |
|-----------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------|
| <b>5 Application Layer</b>              | HTTP, FTP, DNS, SMTP, SSH, DHCP        | Provides network services to end-user applications (web, email, file transfer, etc.) |
| <b>4 Transport Layer</b>                | TCP, UDP                               | Provides end-to-end communication between devices (reliability, ports, segmentation) |
| <b>3 Network Layer (Internet Layer)</b> | IP, ICMP, ARP                          | Logical addressing and routing packets between networks                              |
| <b>2 Data Link Layer</b>                | Ethernet, PPP, Wi-Fi (802.11)          | Frame creation, MAC addressing, error detection at link level                        |
| <b>1 Physical Layer</b>                 | Fiber, Copper cables, Wireless signals | Transmission of raw bits over the physical medium (electrical, optical, or radio)    |

Figure 6.19 An exchange using the TCP/IP model



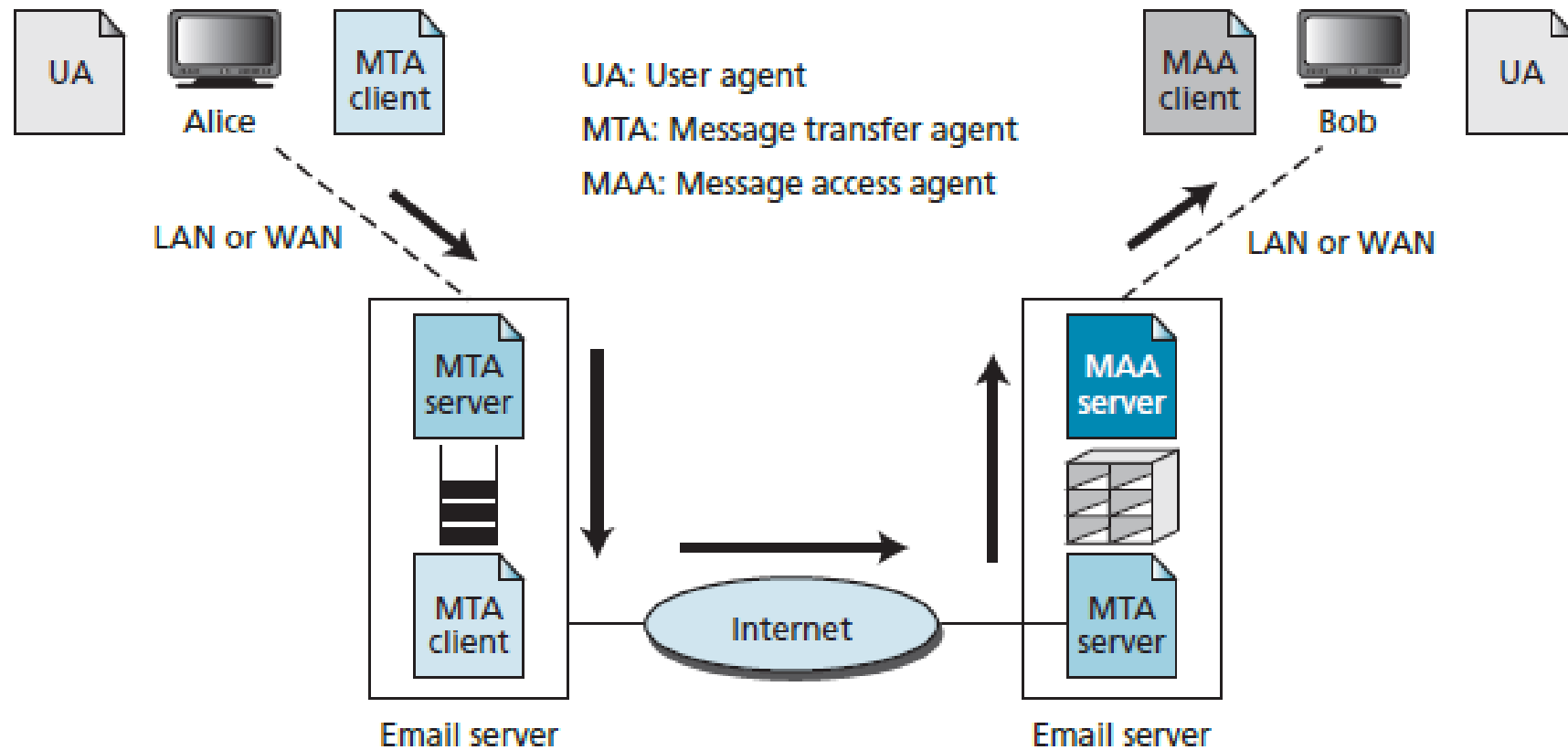
## 6.4 INTERNET APPLICATIONS

- The main task of the Internet is to provide services for users. Among the most popular applications are **electronic mail**, **remote login**, **file transfer**, and accessing the **World Wide Web (WWW)**.

### 6.4.1 Electronic-mail

- Let us first discuss electronic mail (e-mail). Ironically, this first application that we discuss in this section cannot be supported by one client process and one server process. The reason is that e-mail is exchange of messages between two entities.
- Although the sender of the e-mail can be a client program, the receiver cannot be the corresponding server, because that implies that the receiver must let their computer run all the time, as they do not know when an e-mail will arrive. For this purpose, e-mail architecture is designed as shown in Figure 6.20.

**Figure 6.20** *Email architecture*



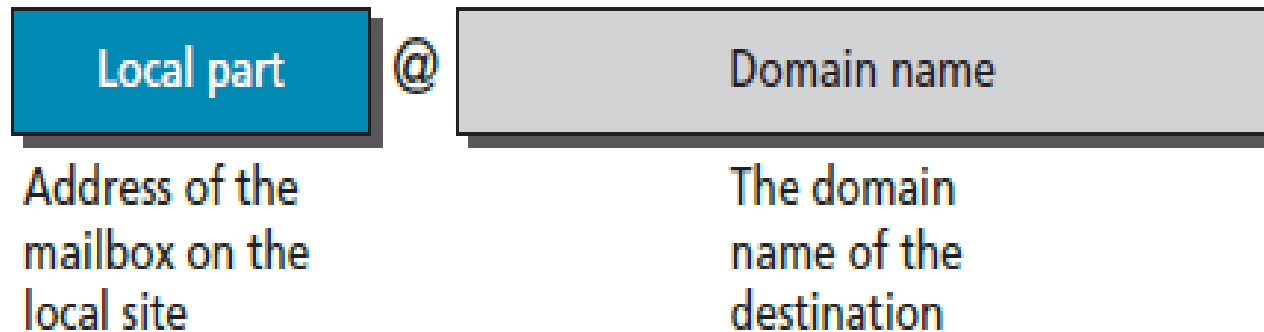
## Mail access protocols

- Stored e-mail remains on the mail server until it is retrieved by the recipient through an access protocol.
- Currently two e-mail access protocols are in common use: **Post Office Protocol, Version 3 (POP3)** and **Internet Mail Access Protocol (IMAP)**.

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Figure 6.21 *Email address composition*

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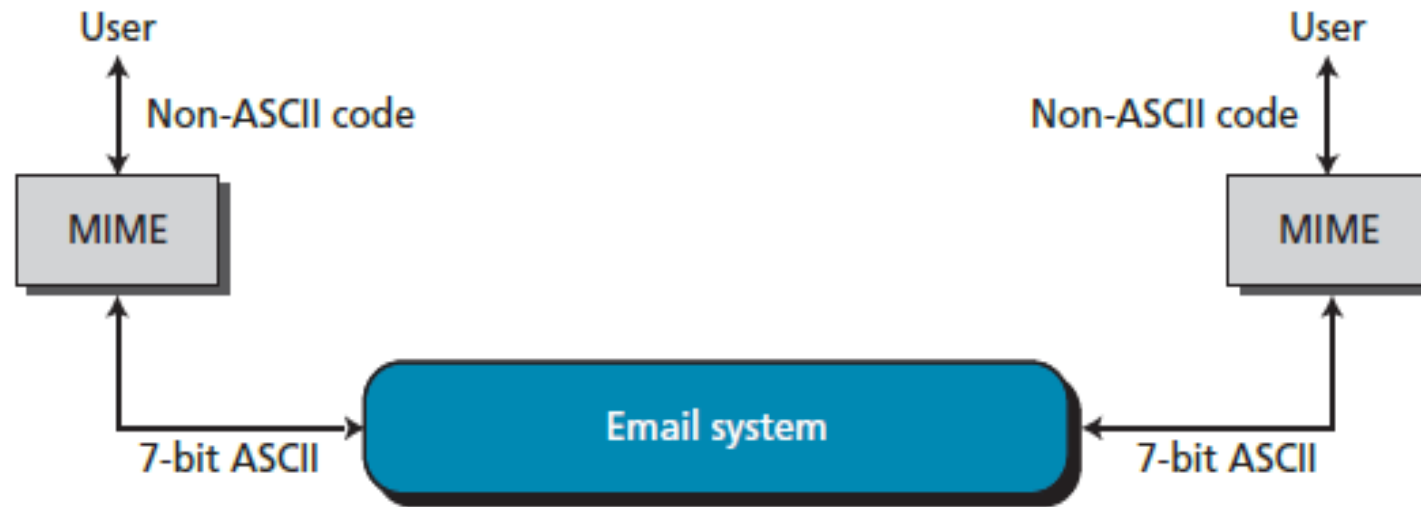




# Multi-purpose Internet Mail Extension (MIME)

- Multipurpose Internet Mail Extension (MIME) is a supplementary protocol that allows non-ASCII data to be sent through SMTP.
- MIME is not an e-mail protocol and cannot replace SMTP, it is only an extension to SMTP.

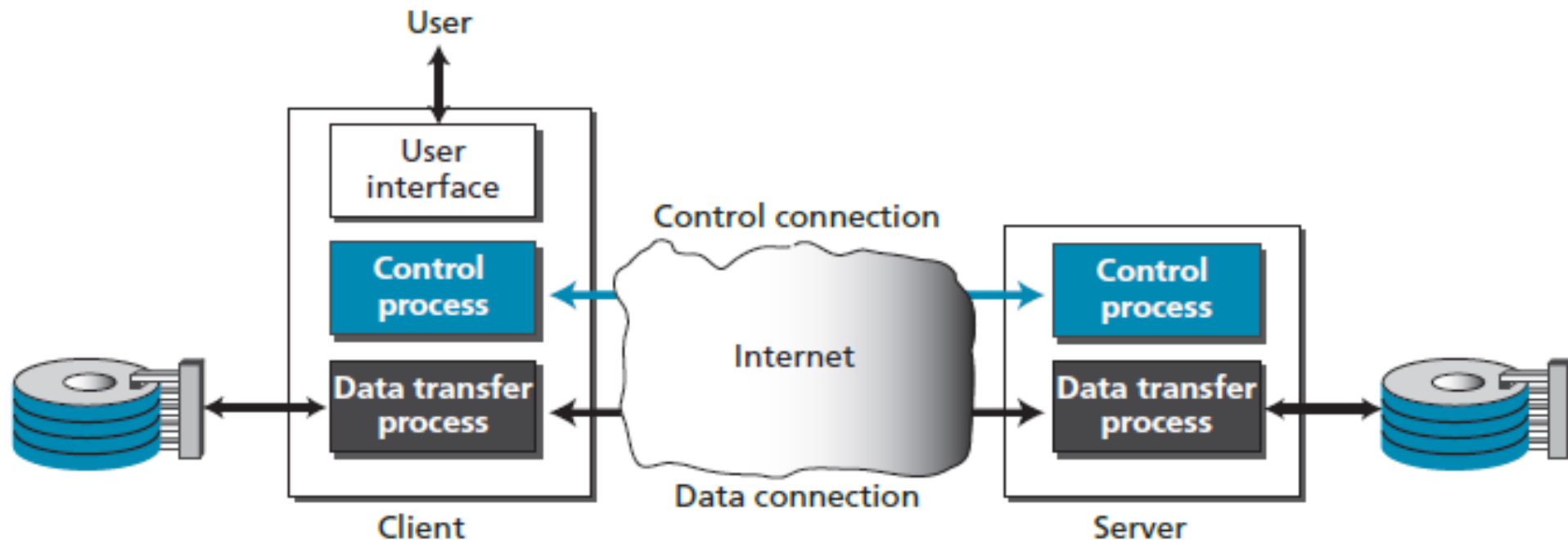
**Figure 6.22** *Multipurpose Internet Mail Extension (MIME)*



## 6.4.2 File Transfer Protocol (FTP)

- **File Transfer Protocol (FTP)** is the standard mechanism for one of the most common tasks on the Internet, copying a file from one computer to another.

Figure 6.23 File Transfer Protocol (FTP)



## 6.4.3 Remote login – TELNET

- The main task of the application layer is to provide services for users. One way to satisfy these demands is to create different client-server application programs for each desired service.
- Programs such as file transfer programs (FTP and SMTP), and so on are already available. However, it would be impossible to write a client-server program for each specific application.
- **TELNET** is a general-purpose client-server program that lets a user access any application program on a remote computer. In other words, it allows the user to log onto a remote computer.
- After logging on, a user can use the services available on the remote computer and transfer the results back to the local computer.

Figure 6.24 *Local login*

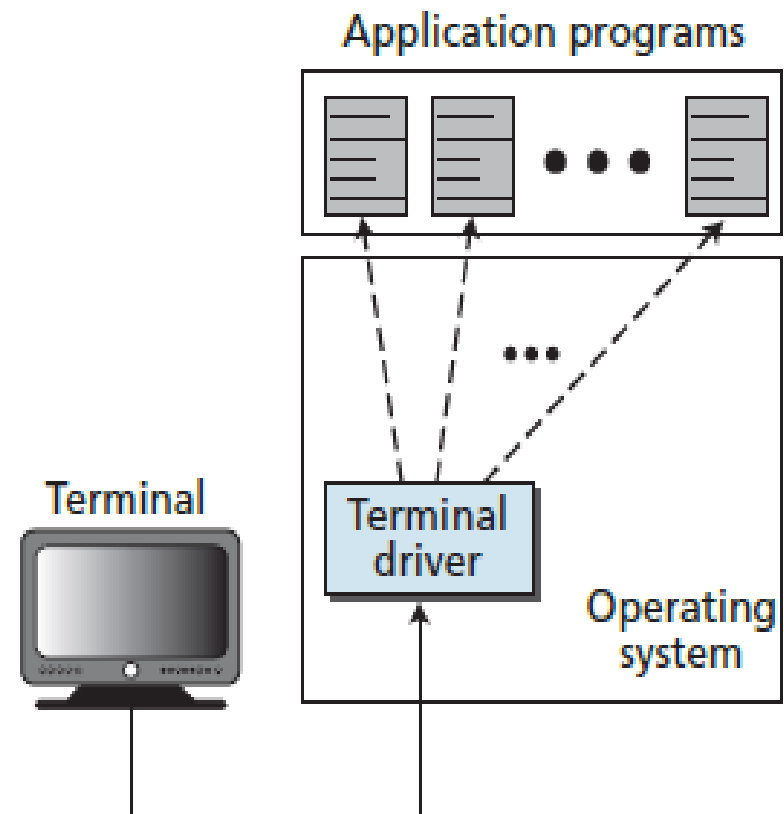
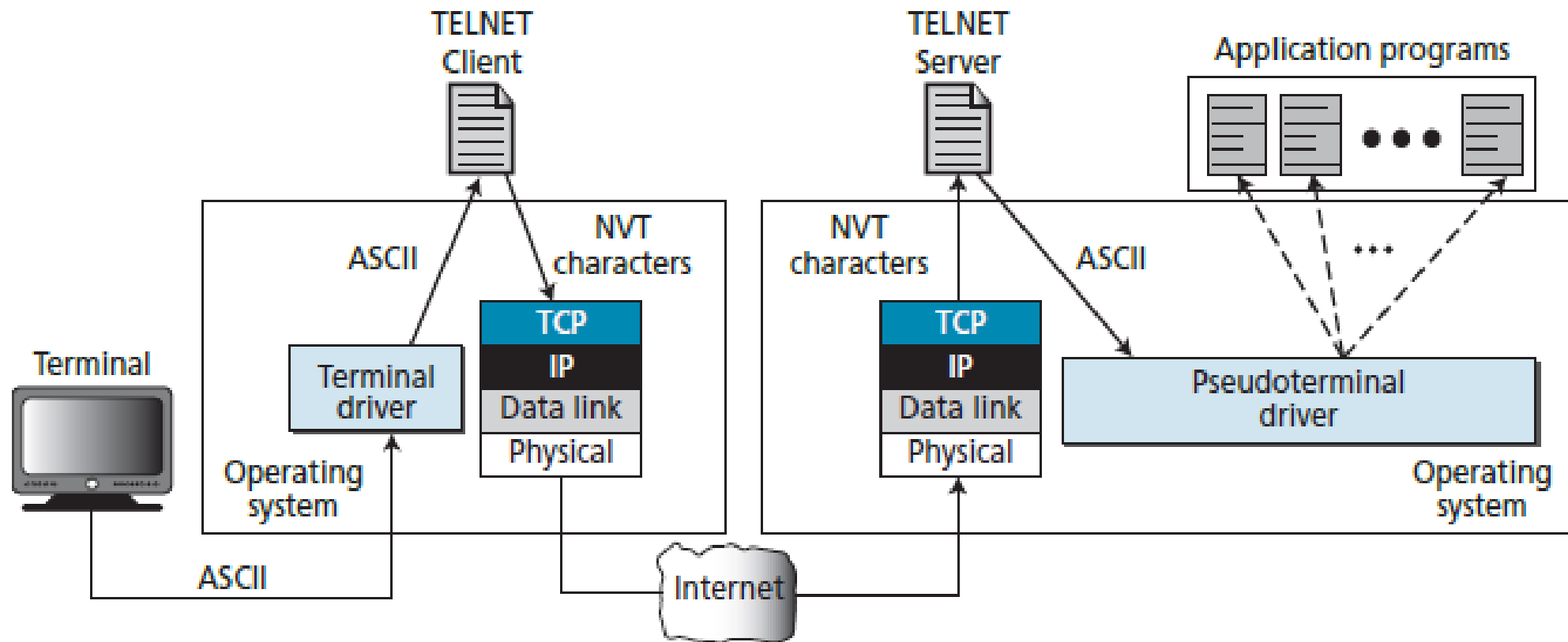


Figure 6.25 Remote login



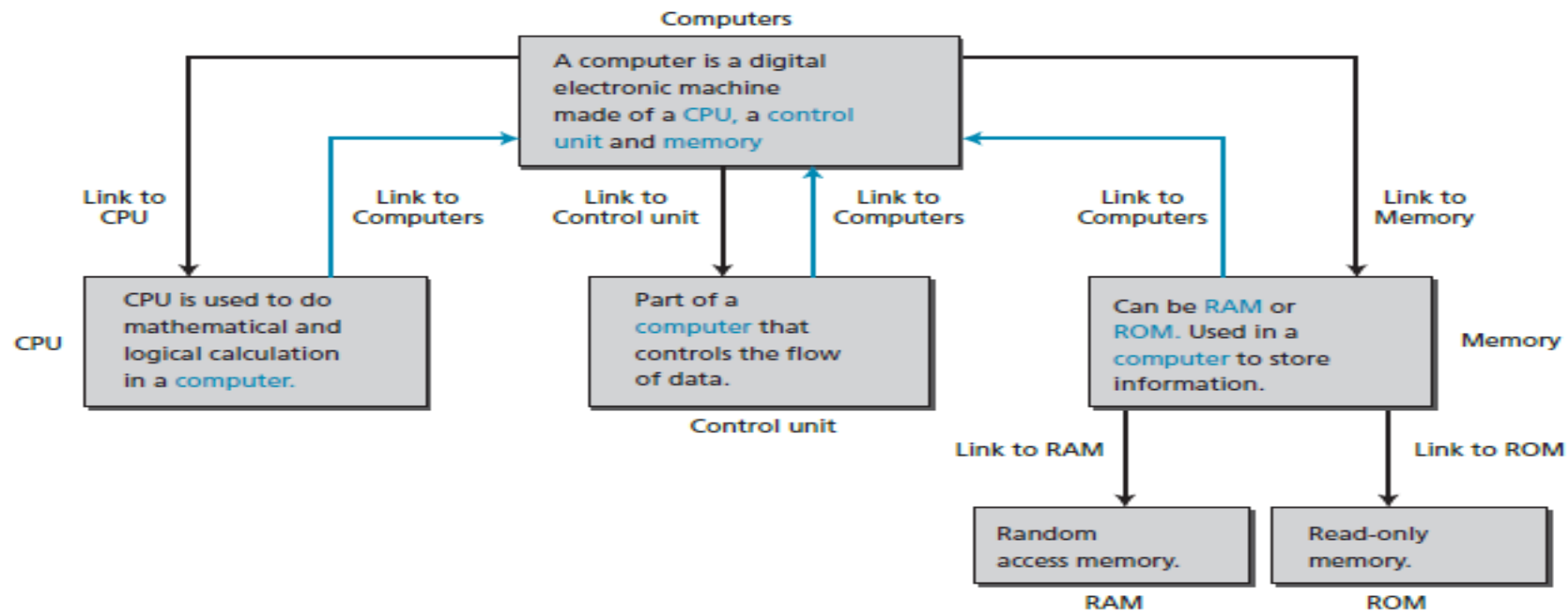
## 6.5 The World Wide Web (WWW)

- The **World Wide Web (WWW)**, or just “the Web”, is a repository of linked information spread all over the world.
- The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet.
- The WWW today is a distributed client-server service in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations, called web sites.

# Hypertext and hypermedia

- The WWW uses the concept of hypertext and hypermedia. In a hypertext environment, information is stored in a set of documents that are connected together using the concept of links.

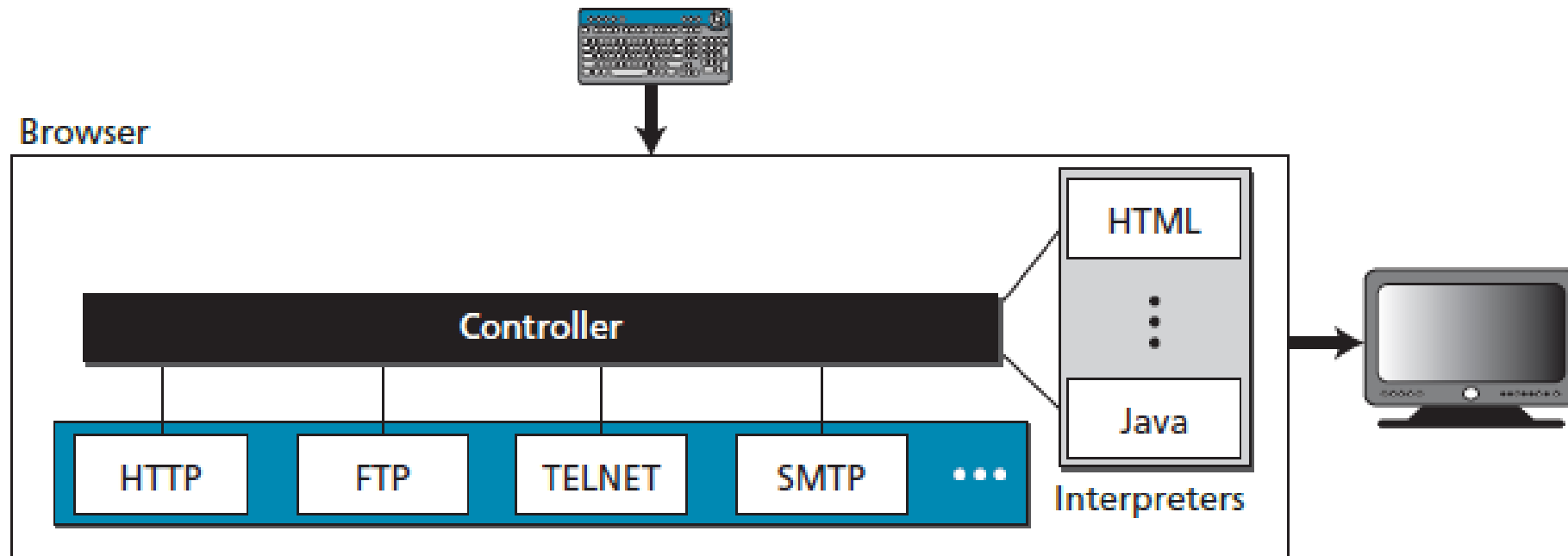
Figure 6.26 Hypertext



# Components of WWW

- To use the WWW we need three components: a **browser**, a **web server**, and a protocol called **Hypertext Transfer Protocol (HTTP)**.

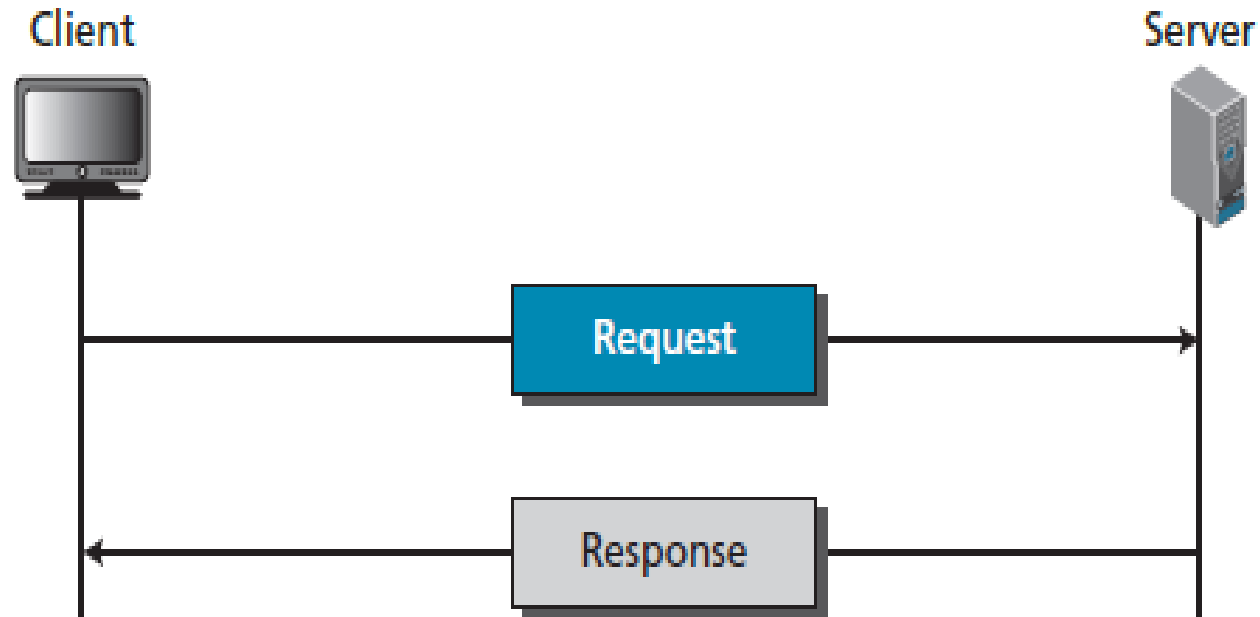
Figure 6.27 Browser structure





- **Hypertext Transfer Protocol (HTTP)** is a protocol used mainly to access data on the World Wide Web.

Figure 6.28 *HTTP transaction*

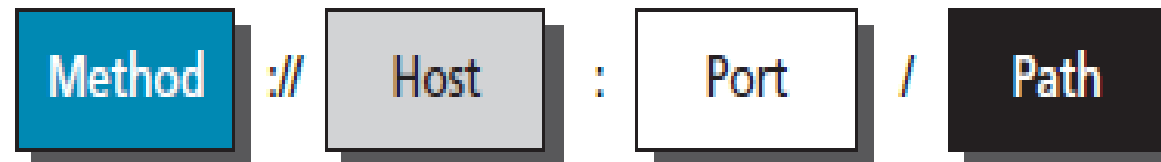


- A client that wants to access a document needs an address. To facilitate the access of documents distributed throughout the world, HTTP uses the concept of locators.
- The **uniform resource locator (URL)** is a standard for specifying any kind of information on the Internet.

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Figure 6.29 *URL structure*

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URL: Uniform Resource Locator

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# Static documents

- The documents on the WWW can be grouped into three broad categories: static, dynamic, and active.
- This categorization is based on the time at which the contents of the document are determined.
- Static documents are fixed-content documents that are created and stored on a server.

# Hypertext Markup Language (HTML) is a language for creating Web pages.

Figure 6.30 *HTML example*

<HTML>

<HEAD>

<TITLE> Small College </TITLE>

</HEAD>

<BODY>

<H1> President: Eve Jones </H1>

<H1> Vice President: Adam Brown </H1>

<H2> Department: Math and Science </H2>

<H3> Courses </H3>

<L1> CS0 </L1>

<L1> CS1 </L1>

<L1> CS2 </L1>

<H2> Department: Business </H2>

<H3> Courses </H3>

<L1> Introduction to Business </L1>

<L1> Business Law </L1>

<L1> Business Mathematics </L1>

</BODY>

</HTML>

Small College

President: Eve Jones  
Vice President: Adam Brown  
Department: Math and Science  
Courses:  
1. CS0  
2. CS1  
3. CS2  
Department: Business  
Courses:  
1. Introduction to Business  
2. Business Law  
3. Business Mathematics

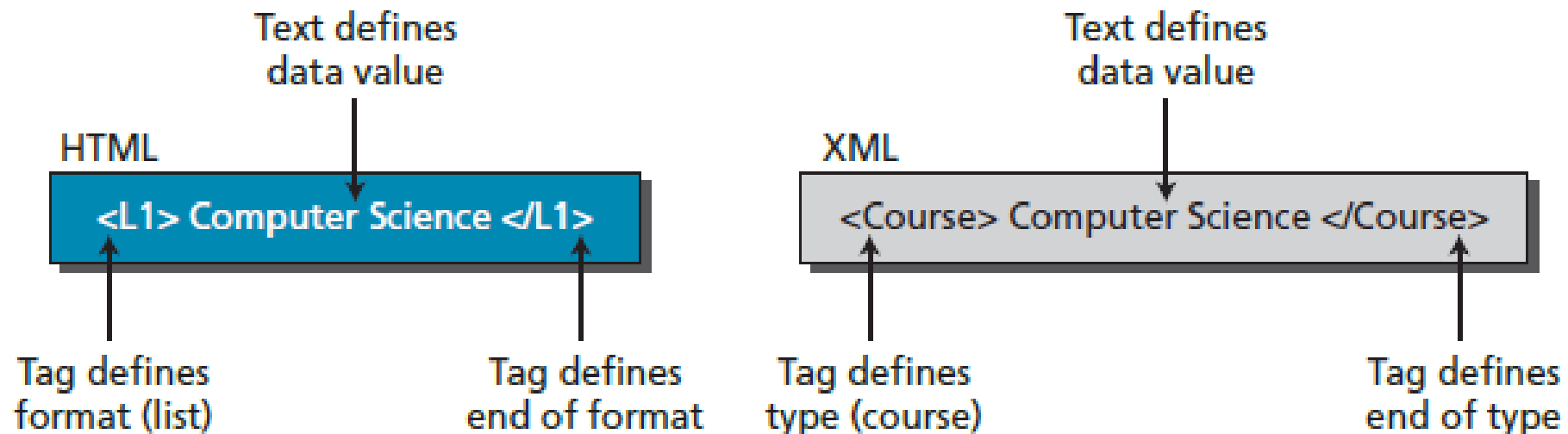
Result

HTML document

6.52

- HTML adds formatting capability to a document, but it does not define the type of data. **Extensible Markup Language (XML)** is a language in which tags can be used to define the content (type) of the text between two tags.

**Figure 6.31** Comparing *HTML* and *XML*



# Dynamic documents

- A dynamic document is created by a Web server whenever a browser requests the document.
- When a request arrives, the Web server runs an application program that creates the dynamic document.
- The server returns the output of the program as a response to the browser that requested the document.

# Active documents

- For many applications we need to be able to run a program or a script at the client site. These are called active documents.
- For example, suppose we want to run a program that creates animated graphics on the screen, or a program that interacts with the user. The program definitely needs to be run on the client computer where the animation or interaction takes place. When a browser requests an active document, the server sends a copy of the document or a script. The document is then run under the control of the client's browser.

## 6.5.1 Other Internet applications

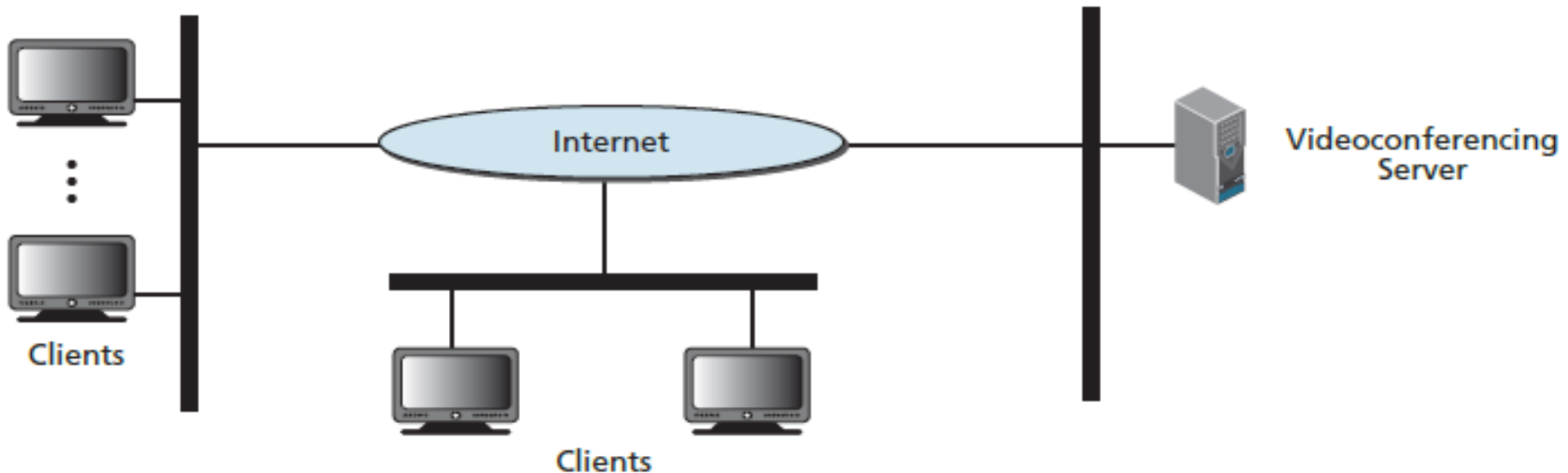
- During the last decades other applications have been developed on the Internet. These applications are not as popular as those discussed above, but they are gaining ground. We discuss a few of them here.



# Videoconferencing

- **Videoconferencing** can eliminate the cost of traveling, and save time and energy, by providing communication between two or more groups of participants or a set of individual participants.

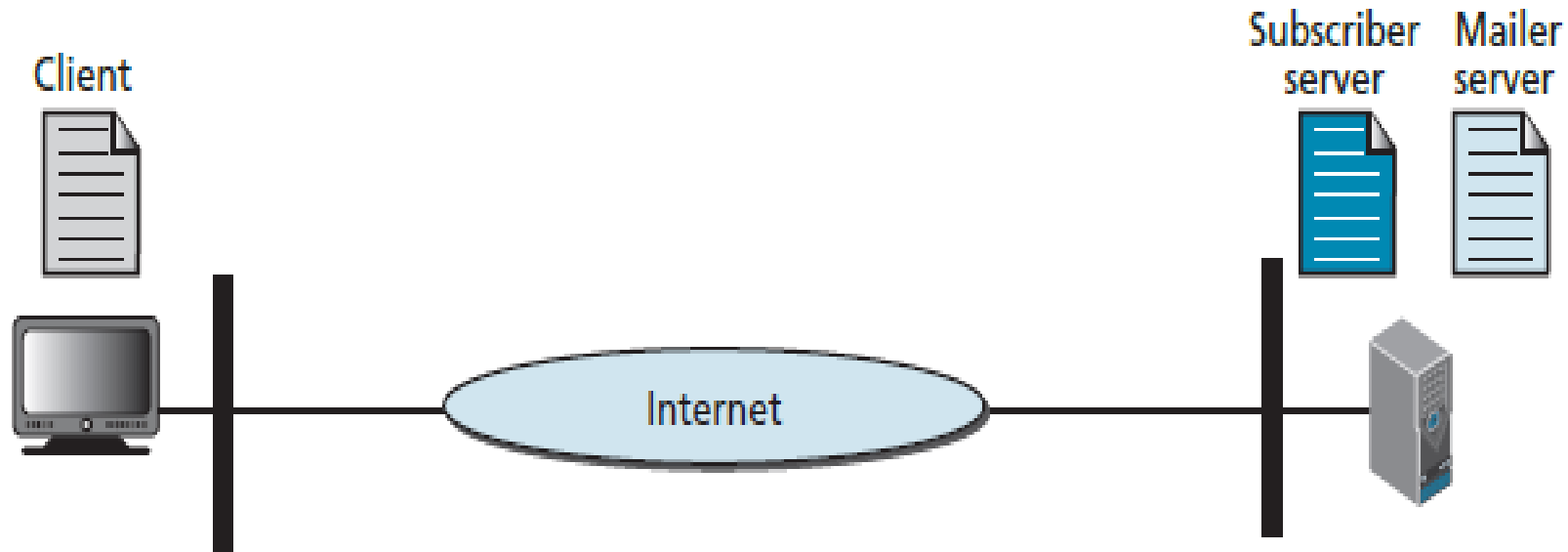
Figure 6.32 Videoconferencing



## Group discussion Listservs

- Another popular class of applications is **listservs**, which allow a group of users to discuss a common topic of interest.

Figure 6.33 *Listsaver organization*



# Chat

- Another popular class of Internet application is **chat**.
- This is a real-time application like videoconferencing, in which two or more parties are involved in an exchange of text and optionally audio and video.
- The two parties can send text to each other, talk to each other (the same way as they might talk on the phone), and even see each other with suitable cameras.

# Exercises

- P183 6.7.3 Problems
  - P.6-2, p6-3, p6-4, P6-4, P6-5, P6-15, P6-16